Chair travel case

ceremonial chairs · build plans · Rev L (12 mm CNC-cut kit, ASA printed fittings)

Case outside	1024 W × 1014 D × 1764 H mm on castors
Inside	1000 × 990 × 1605 mm, lined 20 mm EPE, 20 mm EVA floor
Chair	927 × 920 × 1547 (maker's STEP, 9 Nov), 15–18 mm air gap all round
Build	12 mm black hexa ply walls, top and lid on a 17 mm formply base; CNC-cut panels, glued + pinned, 30 mm single angle, 260 mm deep slide-off lid on a 12 mm hybrid edge split; ASA printed handles, foot rails and brackets
Access	5 butterfly latches; pull the lid straight off forward; chair rolls out forward
Mobility	100 mm castors (100 kg each, 2 braked) bolted to the base at the corners, 6 handles
Weight	about 98 kg empty, 151 kg packed (calculated: weigh the first one)

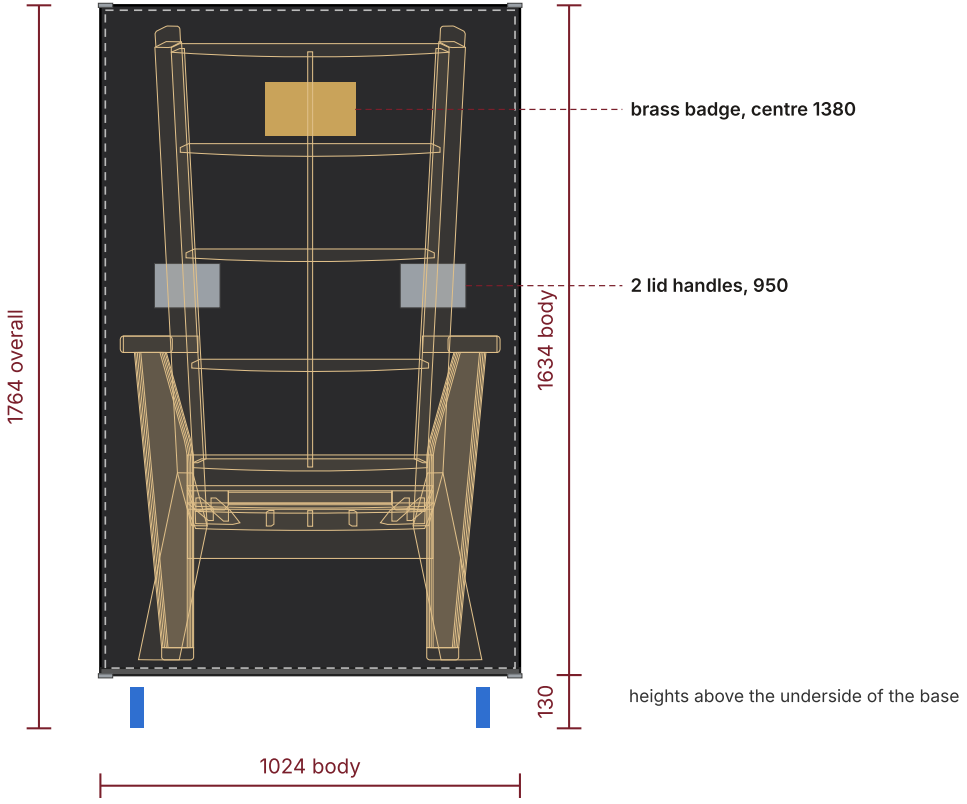
Before the final CNC files go: confirm the real ply thickness, and mate the hybrid joiner on two offcuts to measure its gap (drawn as 8 mm). Set the foot rails on the first finished chair. Print and test one handle first. Chair geometry is maker's STEP model. Badge needs the client' official vector logo.



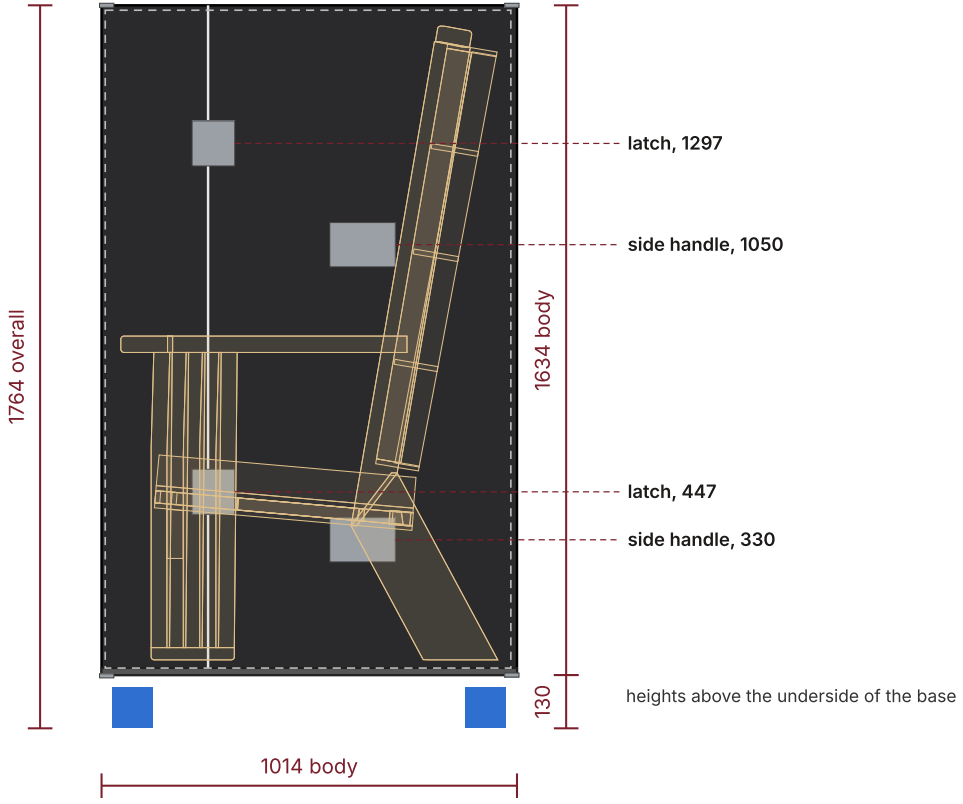


Elevations

FRONT (lid on), chair outlined

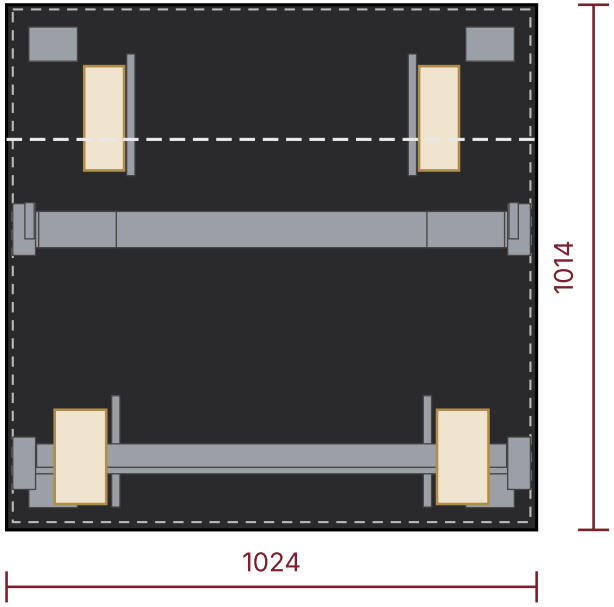


LEFT SIDE, lid at left (split 210 back)

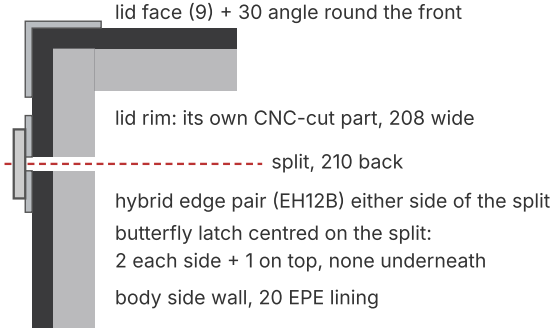


Plan and details

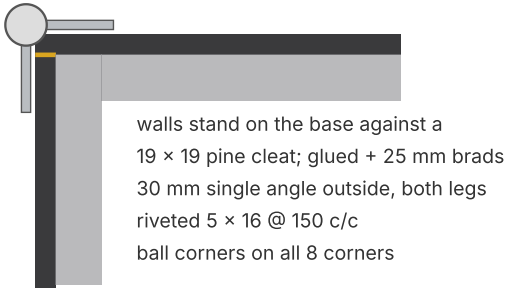
PLAN at floor, lid at top



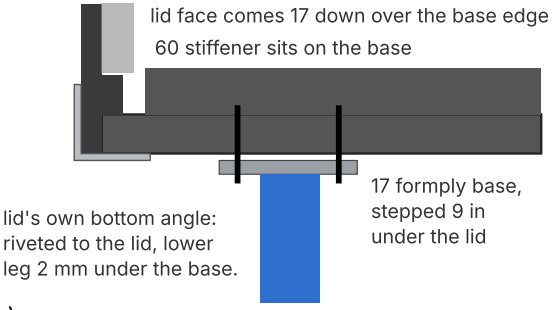
A Lid and split (plan section through a side wall)



B Rear corner (plan section)



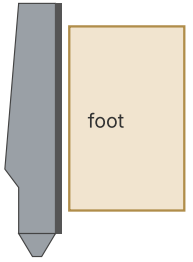
C Lid skirt over the base (section at the front)



Lid off = flat base. Lid pulls straight off.

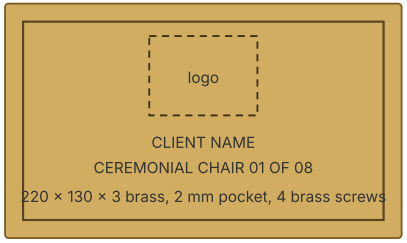
D Foot rail, ASA print (plan)

back rib stops the chair, not these

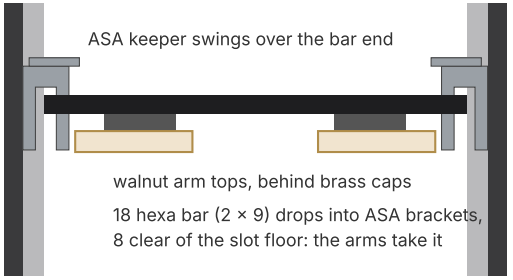


flared end towards the lid

E Brass badge

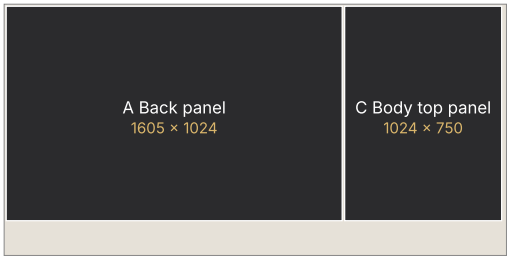


F Hold-down bar (section across case)



Cut list · hexa ply, 4 sheets per case

Sheet 1



Sheet 2



Sheet 3



Sheet 4



CNC parts, one case

Part	Name	Size	Qty
A	Back panel	1024 × 1605	1
B-L	Body side panel LEFT (mirror of B-R)	738 × 1605	1
B-R	Body side panel RIGHT	738 × 1605	1
C	Body top panel	1024 × 750	1
D	Lid face	1000 × 1624	1
E-L	Lid rim side LEFT (mirror of E-R)	256 × 1624	1
E-R	Lid rim side RIGHT	256 × 1624	1
F	Lid rim top	1024 × 256	1
G	Lid bottom stiffener	1000 × 60	2
H	Back rib	976 × 80	1
J	Hold-down bar	976 × 70	1
M	Base	1024 × 1000	1

Panels are CNC cut by the supplier from the Rev L DXF pack (12 mm), with handle openings, bracket bolt holes, latch notches, the badge pocket and castor marks. Left and right are mirror images. The hybrid joiner gap is drawn as 8 mm and the latch notches as 49 (body) and 23 (lid) deep: best estimates from the suppliers' published sizes. Extrusion runs: 30 mm single angle 964, 720, 226, 1574, 695, 1576, 201, 720, 226, 1574, 695, 1576, 201, 964, 964, 964; hybrid joiner 1578, 1578, 974, 1578, 1578, 974.

Build order

1. **Order and collect.** Panels: order the Rev L (12 mm) CNC-cut kits from the chosen panel supplier. Before the final files go, confirm the real sheet thickness and buy one 3 m length of Armor EH12B hybrid edge and one latch so the split gap and latch notches can be checked. Extrusion, latches, ball corners, rivets and foam: Titan AV (ask for uncut 3 m extrusion lengths: five verticals are over 1.5 m). Bolts, screws, glue, contact spray and paint from Bunnings. Start the ASA prints (list on the ASA prints tab) while the panels are being cut.
2. **Print the ASA parts.** Print one set per case in black ASA on the Bambu printers, enclosure closed: 6 recessed handles (face down, 6 walls, 50% infill), 2 front and 2 rear foot rails, 2 rib brackets, 2 bar brackets, 2 bar keepers, 2 front bumpers, 1 label holder. Plus one rivet drill guide for the workshop. All parts fit a 256 mm bed. Print one handle and one keeper first and try them before running the rest.
3. **Check the kit.** Lay out one case's parts against the parts list (A to M). Check the hexa face is on the outside of both side panels and both lid rims: left and right are mirror images. The lid face and rims are 19 mm taller than the body walls: that extra comes down over the edge of the base and 2 mm past it, so the lid angles clear the underside. The front of the base is 2 mm short of the lid face and 1.5 mm clear of each lid side, so the lid slides on without binding. Offer a handle into its cut-out and a latch up to its notches. Mate two EH12B hybrid edge halves on 12 mm offcuts and measure the gap between the two ply edges: the kit is drawn for 8 mm and works from 6 up to about 12 (a bigger gap just lets the lid sit that much further forward).
4. **Prep the parts.** Paint the cut edges of the base black, including the stepped-in front edge that the lid covers. Round over the edges of the back rib (H) and the hold-down bar (J): each is a single 12 mm strip. Glue 20 mm EVA to the front face of the rib and two EVA pads under the bar where it sits on the arms. Cut the three 19 × 19 pine cleats.
5. **Glue up the body and the lid.** Body: glue and screw the pine cleats to the base, 12 mm in from the back and side edges (the wall thickness). Stand the back (A) and the two sides (B-L, B-R, notched edges to the front) on the base against the cleats; glue (PU) and pin with 25 mm brads, then screw through each wall into its cleat every 200 mm with 8g x 25 screws, 6 mm above the base: the 13 mm upper leg of the base angle covers those heads. Glue and pin the top (C) on, notch to the front. Lid: build it ON the body, not on its own, so the two halves cure as one box (the usual case-maker trick is to build a closed box and saw the lid off; our stepped base means it cannot be sawn off, so this does the same job). Stand packers on the split edges of the body, as thick as the gap you measured in step 3 (about 8 mm), then glue and pin the rims (E-L, E-R) to the edges of the face (D), the top strip (F) over all three, and the two 60 mm stiffener strips (G), glued face to face inside the face, 19 mm up from its bottom edge and sitting on top of the base (the stiffener carries the lid; the bottom edges of the lid hang 2 mm below the base). Tape across the split so no glue bridges it. Leave it overnight, then slide the lid off. With a file, ease the inside bottom corner at the back of each lid side, where it meets the rounded corner of the step in the base.
6. **Fit the split frame.** Measure the built box and cut the extrusions to it, not to the drawing (supplier advice: panels and glue-up always move a little). Fit the EH12B hybrid edge pair up both sides and across the top of the split, lid half on the lid and body half on the body, mitred at the two top corners, starting 31 mm above the underside of the case. Cut the hybrid edge away at each latch notch to match. Rivet at 150 mm. Slide the lid on and top sides. Lid = bottom front and bottom sides, front verticals, top front and top sides. The lid's bottom angles are riveted to the lid only; their lower leg runs 2 mm under the base so the lid can't lift or kick out at the bottom, but slides straight off forward. Rivet into the 12 mm ply at 150 mm (4.8 × 19 aluminium rivets, grip 12.7-15.9, with washers; use the printed drill guide). Ball corners on all eight corners: four on the body's rear corners, four on the lid. No flat corners or brace plates. With the lid off, nothing stands up above the base.
8. **Latches and handles.** Five butterfly latches into the pre-cut notches on the split: two each side and one on top, dish half on the body, catch half on the lid. Six ASA recessed handles into the 130 × 75 cut-outs (two each side, two on the lid face): drop in from outside, 6 x M5 × 25 black button-head bolts through the pre-drilled holes, washers and nylocs inside. Bolt the two front bumpers to the inside of the lid face (4 x M5 × 25 each, holes pre-drilled) but leave their EVA pads off until the first-chair fit. Rivet the label holder to the back.
9. **Castors.** The base has the four castor plate centres marked (90 mm in from the sides, 76 mm in from the front and back). Centre each plate on its mark, drill through its holes, and bolt with M8 × 40 cup-head bolts, heads inside (they end up under the floor mat), nylocs under the plate. Braked pair at the lid end.
10. **Brackets for the rib and bar.** The side walls come with the bracket bolt holes pre-drilled. Bolt the two ASA rib brackets (low, near the back, leaning back 22 degrees: the holes set the angle) and the two ASA bar brackets (high, near the front) to the walls with 4 x M5 × 30 button-head bolts each, heads outside. Fit each keeper on an M5 × 30 bolt through the front jaw with a washer either side and a nyloc, snug enough that it clicks into its detent.
11. **Line it.** Cut the foam to the built box. Spray-contact 20 mm EPE on the back, sides and top of the body and on the lid face and rims, cutting round the brackets, bumpers, latch pans and handle pockets. Then the 20 mm EVA floor mat between the linings, with the four foot-rail slots cut out (16 wide; mark them from the first chair, step 12, before gluing the mat down).
12. **First-chair fit.** Drop the rib into its brackets. Roll the first chair in from the front until its rear legs touch the rib foam, centred side to side. Mark the inside edge of each foot on the mat. Lift the chair out, cut a 16 mm slot in the mat 4 mm inside each mark, and screw the four ASA foot rails straight onto the base in the slots (3 × 8g x 25 each), flared end towards the lid, felt on the foot side. Roll the chair back in. Lay the bar into its brackets on the arm tops behind the brass caps and swing both keepers over it. Now fit the front bumper pads: with the lid latched on, measure from each bumper face to the front leg, and glue on EVA to leave about 2 mm (20 mm if the hybrid gap measured 8; add 1 mm of pad for each extra mm of gap). Nothing touches leather or brass.
13. **Badge.** Fit the engraved brass plate into its pre-cut 2 mm pocket on the lid face with four brass countersunk screws.
14. **Test and record.** Roll it, brake it, push it from the high side handle and tilt it 10 degrees. Lid off and on a few times: undo the five latches, hold both lid handles and pull straight towards you; to close, slide it straight back on and do the latches up. Load and unload the chair twice: keepers back, bar out, roll the chair in until it stops on the rib, bar in, keepers over, lid on. The chair is then held all ways: rails sideways, rib and lid bumpers front to back, bar down. Weigh it packed and note it for freight. Photograph it for the handover.

Handling: about 151 kg packed. Tailgate truck or forklift, two people. Brake the castors